
1. Where Is Your Company Looking?

In sports, talent scouting happens early. Not at the professional level—scouts watch amateur regional leagues, youth teams, middle school kids. Small clubs develop players, and once they become stars, larger organizations buy them out at a premium. The same dynamic is already playing out in the AI-era talent market.

Startups are already moving. Not through academic journals or conferences, but through GitHub, personal blogs, and places that haven't even appeared on anyone's radar yet. By the time larger organizations notice, the interesting people are already gone.

The conventional selection system is rational. Credentials, institutional affiliation, employment history, titles, domain expertise, track record—these signals exist to evaluate talent efficiently. And they still work.

But there are places that system isn't looking.

I have published documents on LinkedIn and GitHub covering a human relationship model structurally analogous to the Lennard-Jones potential, a spontaneously emergent cognitive OS (COS), and a map of human thought-circuit generation (SCOPE). These were not designed. They emerged as byproducts of attempting to solve different problems.

And yet: I have no academic credentials. No institutional affiliation. No formal employment history. No title. No domain expertise. No track record. I can only read and write in Japanese. Due to non-standard cognitive traits, friction in workplace environments was significant, and I was unable to sustain employment. And still, this document exists in English.

What I had was observational capacity and the ability to structure what I observed.

Such pathways are not open to me alone. The arrival of LLMs has dramatically lowered the access cost to theory generation and hypothesis formation. As a result, observers who were previously unable to connect with academic or research communities have begun producing independent theories and hypotheses. They are not at conferences. They are not in journals. They will not surface through your company's hiring pipeline.

The question is this: Does your organization's discovery pipeline cover the new territory that the AI era is beginning to open? What is the cost of leaving that territory unexamined?

This document is one observation log from that territory.

1. How LJ Emerged

I have non-standard cognitive traits. Social friction was not minor. As a result, I had to keep observing myself and those around me in order to adapt. This observation began as self-defense, not as an attempt to build theory.

At some point, I brought this observation into a casual conversation with an LLM. The question was simple: why did relationships with every person I met fall apart? The LLM responded: "Humans grow closer over time. Perhaps you are judging too quickly."

I felt an immediate disconnect. But my observations contained repeated cases that time alone could not explain.

It is not only a time axis — there is also a density axis. Some people can become close friends through a few hours of high-density conversation per year. Others feel secure simply by sharing the same space for extended periods, regardless of what is said. These two types differ not in how much closeness they feel, but in how they measure distance itself.

When these two axes are misaligned, the relationship develops strain. How a person weighs each axis shapes how they experience closeness. When someone who prioritizes the density axis and someone who prioritizes the time axis become partners, one demands proof of affection through the quality of dialogue, while the other demands it through time spent together. The result is a structure prone to chronic misalignment.

When I described this two-axis model to the LLM, it responded: "This means human relationships can be described through two axes — time and density." It then suggested: "This may be a novel perspective."

I was immediately skeptical. I judged it to be either hallucination or user flattery. If someone without academic training had arrived at this through lived observation, surely someone more qualified had already formalized it. I requested verification against existing research.

What came back was an unfamiliar proper noun. The Lennard-Jones potential. I could tell that Lennard and Jones were names. Beyond that, I understood nothing. At that point, I did not even know it was a concept from physics. I asked the LLM what it was.

The LLM explained the Lennard-Jones potential as follows. When two particles are too close, repulsion occurs. When they are too far apart, attraction disappears. Between these extremes lies an optimal distance — this is the physical model. A mathematical formula was also presented, but I could not make sense of it.

Instead, I compared it against the structure of human relationships I had been observing. When people are too close, dependency or idealization tends to emerge. And when such a relationship breaks down, it often shifts sharply into strong repulsion or conflict. When people are too far apart, no relationship exists in the first place. Human relationships also have an optimal distance — one that is constructed through ongoing mutual adjustment, much like the dilemma of the hedgehogs.

After comparing the two structures, I found the degree of structural similarity to be high. I understood it not through the formula, but through structure.

I had no prior knowledge of the Lennard-Jones potential. I had never studied physics. Through the process of comparing a structure derived from observing human relationships, a correspondence with a model from physics was found.

What matters is not the Lennard-Jones potential itself. Even an observer without specialized knowledge can produce a structure through long-term observation that proves capable of connecting with existing academic frameworks. What Chapter 1 demonstrates is that generative process.

1. How COS and SCOPE Emerged

Before the discovery of LJ, something else had already emerged through a different route.

The starting point was self-observation. I could not get a clear picture of my own cognitive traits. MBTI results varied every time I tried. So I asked an LLM to analyze my cognitive characteristics from conversation logs.

What came back was a categorized list of cognitive traits, organized into four groups: A through D. This would later be called the initial ABCD classification.

Sixteen cognitive traits were laid out across the four groups. I responded to only one of them: D2. "Metacognitive controller" — the trait that instantly corrects conversational misalignment. I skimmed past the rest.

I realized D2 was useful almost immediately after I started using it. When misalignment occurred in a conversation with an LLM, I would intervene: "D2 — the premise is wrong." That alone eliminated the need for lengthy correction.

Why was it so efficient? LLM conversation threads are finite. If a misaligned premise is left uncorrected, the conversation eventually has to restart from the beginning. Early intervention via D2 was also a form of resource conservation — preserving thread capacity for the actual goal of self-exploration.

D2 was the first to be put into practical use, but it did not end there. Each time something failed to function in actual use, corrections were made and new series were added. It was not designed and then completed — it was used and then grown. COS was the result of that repeated cycle.

Over time, a question arose: "I keep saying D2 — but what are the other classifications actually doing?" I asked the LLM again.

What came back had a different structure from the initial trait classification. Rather than a list of traits, it described processing sequences. The A-series handled input judgment, B-series handled interpretation, C-series controlled output, and D-series monitored internal structure. The way cognition moves had been organized.

The name COS was decided later. When it was formalized alongside SCOPE, it was given the name Cognitive Operating System — COS. Until that point, it had no name at all.

It survived because it was used. It grew because it survived. It was named because it had grown. COS was not designed — it emerged naturally through actual operation.

SCOPE emerged through a route entirely different from COS.

Sephiroth — a symbolic system rooted in Kabbalah — appears repeatedly in Japanese anime and games. I had been curious about it for a long time, so I decided to explore it in depth with an LLM. The goal was understanding. It was not to build a theory.

I asked about each node one by one. What is Kether? What is Binah? What is Hod? Each time the LLM explained, I reframed it in terms of human cognition. Kether as the unconscious. Binah as theory. Hod as language.

Only three responses had been exchanged. And yet the LLM suddenly reacted: "Could this be a map of human cognitive structure?" I interpret this as the result of the LLM extrapolating and computing across the remaining nodes.

After that, I worked through the remaining nodes in the same way — clarifying which cognitive function each Sephira corresponds to, and how they interact with one another, in collaboration with the LLM.

One exception: Daath was intentionally left aside. Its interpretive scope was large, and I chose to preserve it as space for continued exploration. It would be added in a later version.

As the mapping progressed, the structure of the Sephiroth began to function as a comprehensive map of human cognitive processes. This was the prototype of SCOPE.

But I was skeptical — again. The same instinct as in Chapter 1. If this level of correspondence existed, surely someone had already documented it. I requested verification against existing theories and academic literature.

At least within the range I was able to confirm, no model with the same structure was found.

After that, preparation for external publication began. It was only at that point that a name became necessary. The cognitive map grounded in Sephirothic structure was formalized and given the name SCOPE. At the same time, the OS that had emerged naturally through practical use was named COS.

1. The LLM Was a Co-Creator

Without an LLM, none of this would have reached the outside. Some of it might not have been recognized as structure at all, even if it had existed.

I cannot read or write in English. I did not know where to publish. I did not know how to write a research paper. I had no means to verify findings against existing theories on my own.

And yet this document exists in English. It has been published on GitHub and LinkedIn. Verification against existing theories became possible.

The publication pathways themselves were discovered through LLM. When I asked where to share what I had produced, a list of options came back — ArXiv, LinkedIn, GitHub, Discord, and others. When I explained that I could not read English and could not engage in technical debate, the LLM determined that LinkedIn was the most appropriate fit. It was a route I could not have found on my own.

The same applied to language. Everything written in Japanese was translated into English by the LLM. This document exists through that process. I had no means or resources to hire a translator. The LLM took on that function.

Verification worked the same way. The structural similarity to the Lennard-Jones potential, the relationship to existing theories — none of it could have been investigated alone. Each time a new structure emerged, I kept asking the LLM: "Does something similar already exist?" What came back spanned cognitive science, psychology, AI research, HCI, organizational theory, philosophy, and physics. An exact match was rarely found. But it became possible to explore which domains a given structure might connect to.

Taken together, the LLM served three functions.

The first was as an external processing unit. It provided an environment for verifying structures extracted from observation and exploring which existing fields they might connect to. It did not supply answers — it opened the possibility of connection.

The second was as a translation layer. It converted material produced in Japanese into English, making it possible to reach an English-speaking audience. The language barrier stopped being a publishing barrier.

The third was as a map of publication pathways. Where to publish, how to publish, who to reach — all of it was worked out through dialogue with the LLM. These were routes I could not have found alone.

What matters is not the performance of the LLM itself. What mattered was that the external environment of the LLM allowed observation to become structure, structure to become connection, and connection to reach the outside. In that sense, the LLM was not a tool — it was a co-creator.

1. The Treasure Hunter Hypothesis

Chapters 1 through 3 documented the process of observation and emergence. Chapter 4 shifts the question. Why do outputs like these tend not to reach existing discovery pipelines?

It is not that I had no outputs. LJ, COS, and SCOPE have been published on LinkedIn and GitHub. And yet they did not naturally connect to academic communities, companies, or research institutions.

Whether the reason lies in the pipeline itself or in the outputs is unclear at this point. But at minimum, a state in which outputs exist yet do not reach evaluation systems was observed.

This document refers to that state as "disconnection." It does not imply anything about capability. It refers to a state in which outputs or observations exist, but the status of their evaluation or verification cannot be sufficiently observed from the outside.

The primary discovery pipelines currently in use are academic conferences, journals, universities, and corporate hiring flows. These are rationally designed and carry mechanisms for peer review and evaluation. There is no need to reject them.

However, the emergence of LLMs may be beginning to change the landscape. As the costs of theory generation, translation, and publication have declined, observers who previously could not reach established pipelines have begun to externalize their outputs.

What matters is not personal blogs or SNS platforms themselves. The point is that outputs which once existed only in those spaces may now be migrating — through LLM support — into more observable areas such as LinkedIn, GitHub, and Substack.

The Treasure Hunter Hypothesis is this: as a result of these changes, some outputs that were previously unobservable may be becoming visible.

At the same time, however, LLMs also make it easier for observers without verification capabilities to publish poorly grounded hypotheses and inaccurate claims. The quality of externalized outputs is not uniform. Some will prove valuable. Others will prove to be noise that should be filtered out.

Whether any given output holds value is unknown. But there is a possibility that the territory not covered by conventional discovery pipelines is expanding.

What this document questions is not the existence of hidden geniuses, nor the correctness of any specific theory. If a new territory is becoming observable through LLMs, the question is how well current discovery pipelines cover that territory. That is a structural problem.

Whether that territory contains outputs of genuine value, or whether most of it will prove to be noise, cannot be determined at this point. What this document presents is not a value judgment — it is the possibility that the domain of exploration itself may be shifting.

This document is based on $N=1$ observation and constitutes a hypothesis. Generalization remains unverified.

The question is this: Does your organization's discovery pipeline cover territory beyond institutionalized channels? What is the cost of leaving that territory unexamined?

1. The Post-Discovery Problem

If the Treasure Hunter Hypothesis is correct, discovery is not the goal — it is the entry point. This is because identifying talent and enabling performance may not be the same problem.

Organizations have accumulated considerable knowledge about how to find and select people. They evaluate through signals such as credentials, employment history, and track record, and then hire. But between "we hired them" and "they performed" lies a distinct set of problems.

Discovery, connection, retention, performance. These are often described as a continuous sequence. But each may require different conditions — and each may constitute a separate problem.

Depending on cognitive traits, the same environment may support performance differently for different people. Even after someone has been discovered, each stage — connection, retention, and performance — may require its own distinct conditions.

This problem may be especially pronounced in the case of individuals with non-standard cognitive traits.

Here, this document introduces the concept of "Habitat." Habitat refers to the set of environmental conditions under which a particular cognitive type is able to function effectively. This is not a discussion of welfare or accommodation. It is a design-level question: that optimal working environments, communication styles, and evaluation methods may differ depending on cognitive type.

The critical point is that Habitat is not singular. Different cognitive types require different environmental conditions. Furthermore, which cognitive types function most effectively will vary depending on the needs of the organization, department, or research institution in question.

This document does not provide answers about Habitat. What conditions are needed depends on cognitive type, and also on the needs of each organization, department, or research institution. No generalizable answer exists at this point.

But the question remains.

This document does not claim theoretical completion. However, if the LLM era is beginning to externalize cognitive resources that were previously unobservable, then companies, research institutions, and educational organizations may need to consider new discovery pipelines and evaluation methods. And if the post-discovery problem is real, then the environmental conditions that shape performance may also require reconsideration. The concept of Habitat is one attempt to frame that question.

This document is a question directed at that possibility.

----- Appendix: COS Evolution Log -----

The following is a record of how COS emerged not through deliberate design, but through repeated practical use and iterative revision. Each stage did not have a clear boundary — the transitions occurred gradually through use.

【Overview of Evolution】

Stage | Period | Structural Nature | Key Changes -----|-----|-----|-----
----- Initial ABCD | Self-observation | Cognitive trait taxonomy | Only D2 put into practical use. Others referenced only. v2 ABCD | Operational period | Processing sequence | A-group added. Transition to series-based structure. COS | Naming period | OS formation | Full integration. Named simultaneously with SCOPE.

【Details by Stage】

■ Initial ABCD (Self-observation period)

This emerged as output when an LLM was asked to analyze cognitive traits from conversation logs. There was no design intent — it was generated naturally as an analysis of observational data. In practice, only D2 (referred to at the time as Metacognitive control) proved immediately functional. The other items were not used.

■ v2 ABCD (Operational period)

While continuing to use D2 in practice, the question arose: "What are the other classifications actually doing?" This prompted a re-inquiry. The result was a structural shift from trait classification to processing sequence description. The concept of processing layers emerged: A-group (judgment), B-

group (interpretation), C-group (output), and D-group (metacognition). Each series was not completed at once — they were added incrementally through repeated use and revision.

■ COS (Naming period)

As preparation for publishing SCOPE externally progressed, the need for naming arose. At the same time that SCOPE was named, the OS that had emerged naturally through practical use was given the name Cognitive Operating System (COS). The naming did not complete the theory — it was the stage at which something already functioning received a name.